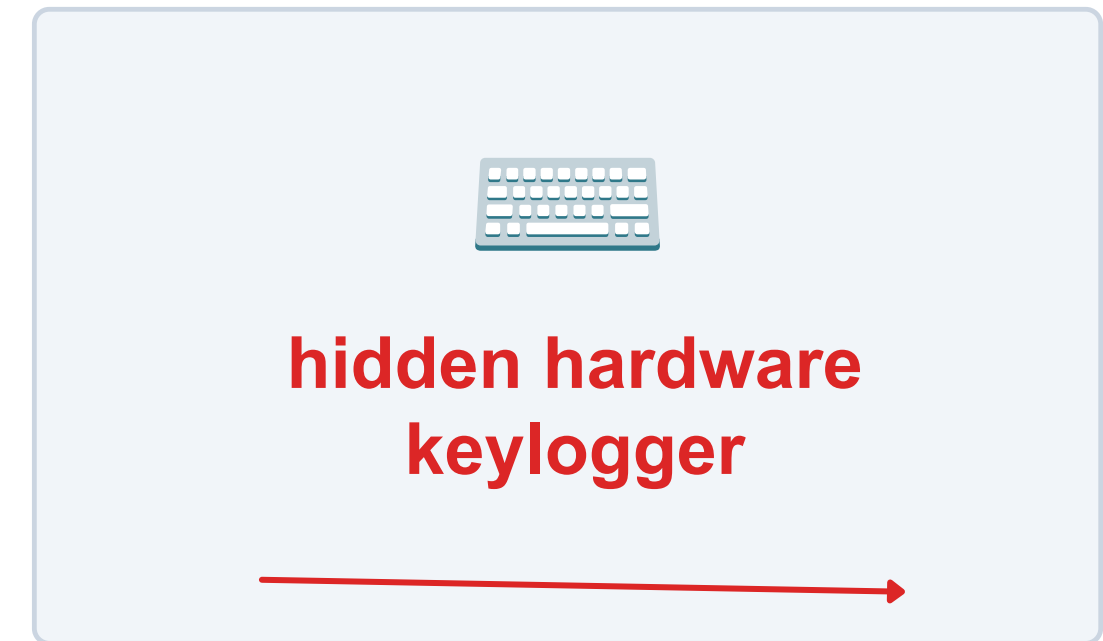


HARDWARE KEYLOGGER IN THE ACCOUNTING OFFICE

Analysis of Physical Access, Workstation Security, and
Authentication Vulnerabilities

Case Study Presentation

20-minute limit



credentials captured

Incident Snapshot

An unauthorized person physically accessed an accounting workstation and secretly attached a hardware keylogger.

Because inspections and physical-access controls were weak, the device stayed unnoticed.

Plain-text passwords and weak authentication made the impact more serious.

Accounting Office

Accounting Office
Primary target

Keylogger

Keylogger
Main attack

Physical + Logical

Physical + Logical
Main weakness

Critical

Critical
Overall risk

Simple idea: the attacker did not need to “hack” first. The attacker used physical access to capture what users typed, then weak software controls made stolen credentials more dangerous.

Important Assets at Risk

1

Hardware

- Desktops & workstations
- Keyboards / input devices
- Servers & storage
- Network equipment

2

Software

- Accounting system
- Operating systems
- Database system
- Authentication/login

3

Information

- Usernames & passwords
- Payroll records
- Employee personal info
- Financial reports

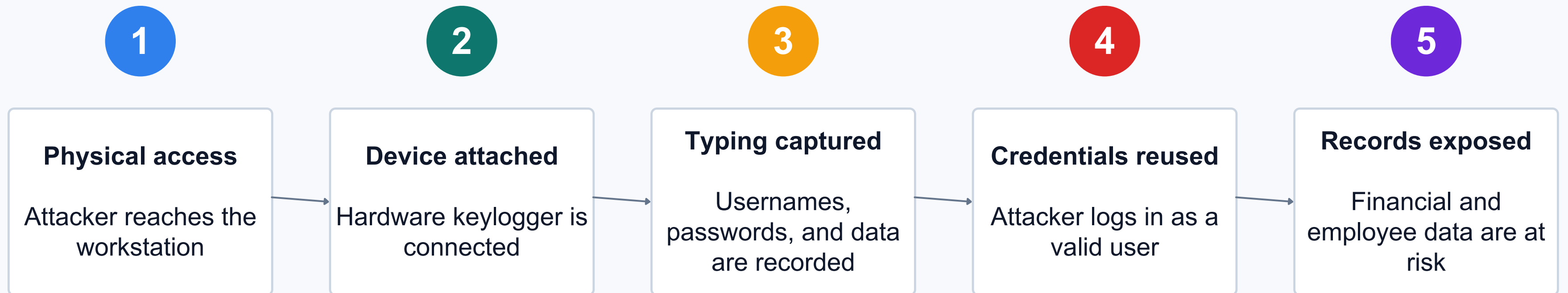
4

Services

- User authentication
- Payroll processing
- Financial reporting
- Backup & recovery

Most sensitive: user credentials, financial information, and accounting records — because exposure or modification can cause fraud, privacy issues, financial loss, and loss of trust.

How the Attack Happens



Key point: a hardware keylogger attacks the input path. Even a strong password can be stolen if the device captures it before the system protects it.

Main Vulnerabilities

Hardware Vulnerabilities:

- Unauthorized physical access
- No regular workstation inspection
 - Unrestricted ports/peripherals
 - Poor hardware inventory
- Workstations accessible to visitors

Physical weakness opened the door

Software Vulnerabilities:

- Plain-text password storage
- No multi-factor authentication
- Weak access-control practices
- Poor suspicious-login monitoring
- Insufficient logging and alerting

Software weakness increased the damage

CIA Triad Impact



Confidentiality
Impact: Critical

Credentials, payroll data, financial information, and employee details can be exposed.



Integrity
Impact: High

Stolen credentials can be used to modify financial transactions, payroll records, and user accounts.



Availability
Impact: Medium–High

Privileged stolen accounts may disable accounts, alter settings, or interrupt accounting services.

The biggest direct risks are Confidentiality and Integrity because the keylogger can steal credentials and those credentials can be used to change records.

Risk Assessment Summary

Vulnerability	Likelihood	Impact	Risk
Hardware keylogger attached	High	Critical	Critical
Plain-text password storage	High	Critical	Critical
Unauthorized physical access	High	High	High
Password-only authentication	High	High	High
Weak workstation locking	High	High	High
Insufficient monitoring	Medium	High	High

Overall Risk Rating: CRITICAL
This rating is justified because accounting systems contain sensitive records and the incident combines physical compromise with credential exposure.

Recommended Security Controls

Hardware

- Inspect workstations
- Restrict ports
- Maintain inventory
- Use tamper-evident controls

Physical

- Restrict office access
- Use ID/access control
- Escort visitors
- Secure unattended areas

Software

- Hash passwords
- Implement MFA
- Log important events
- Patch systems

Logical/Admin

- RBAC + least privilege
- Account reviews
- Security policies
- Awareness training

Defense in depth: do not depend on one control only. Use layers so that if one control fails, another control can still reduce damage.

Prevention Strategy

Secure Hardware Implementation

Treat accounting computers as high-value assets. Restrict entry, inspect devices, control peripherals, monitor the area, and document hardware changes.

Secure Software Development Lifecycle

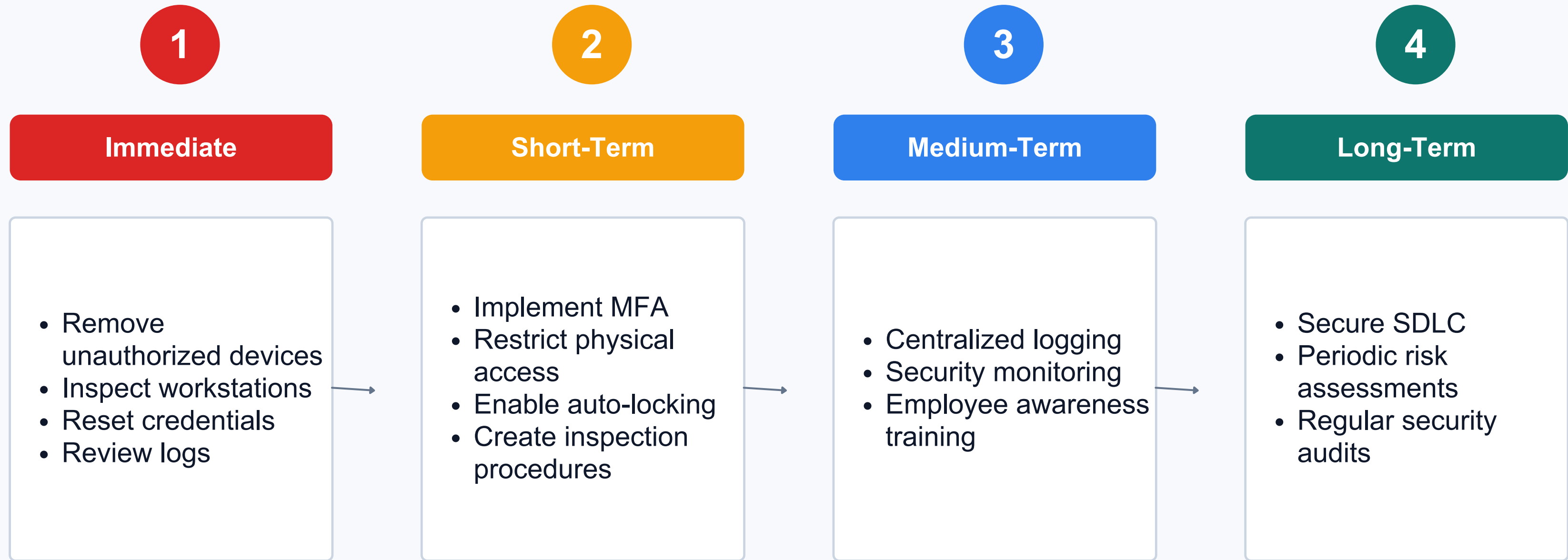
Build security from requirements to maintenance: authentication, authorization, password storage, logging, testing, secure deployment, and updates.

Prevent unauthorized device installation

Reduce account takeover risk

Best prevention = secure physical access + secure authentication + secure password storage + continuous monitoring.

Security Improvement Roadmap



Goal: eliminate current compromise, prevent repeated attacks, improve detection, and reduce future hardware/software vulnerabilities.

Conclusion and References

Conclusion

The incident shows that security is not only a software issue. A keylogger became dangerous because physical access, workstation inspection, authentication, password storage, and monitoring were all weak. The organization should apply defense in depth to protect people, equipment, credentials, applications, and information.

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End of Presentation